

D0 High-Level Design

Carter Smith, Alex Watts

Title, goal statement, and conventions.

Title: Steganography on JPEG vs. PNG

Goal: Find out if hidden messages are harder to detect in JPEG than in PNG.

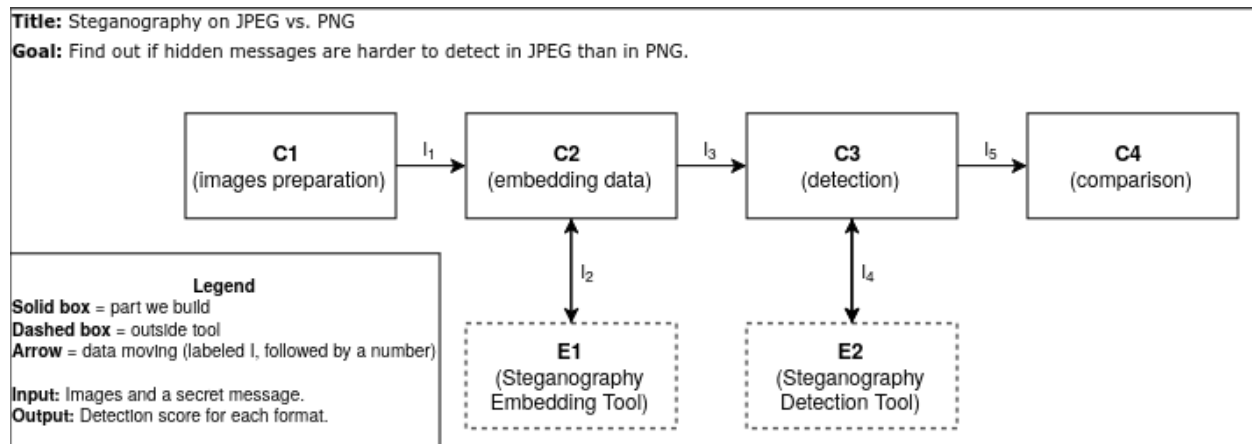
Input: Images and a secret message.

Output: Detection score for each format.

Conventions:

- Solid box = part we build
- Dashed box = outside tool
- Arrow = data moving (labeled I, followed by a number)

Block diagram (the D0 diagram).



Component responsibility table.

Component	Responsibility	Interface In	Interface out	Owner
C1 (images preparation)	Prepare images in both file formats.	<u>NONE</u>	I1	Carter
C2 (data embedding)	Hide data in images using E1.	I1 and I2	I2 and I3	Carter
C3 (detection)	Check the images for hidden data using E2.	I3 and I4	I4 and I5	Alex
C4 (comparison)	Compare each file format's detection rate.	I5	<u>NONE</u>	Alex

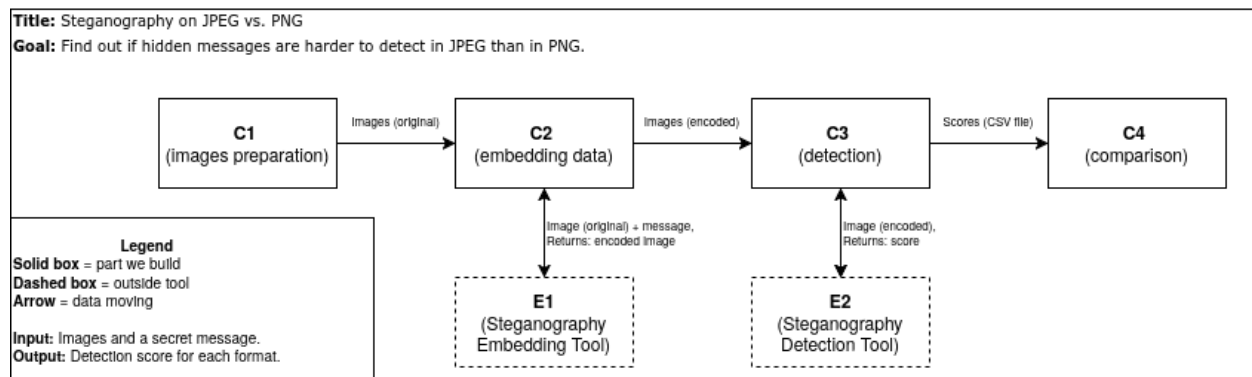
Interface specification table.

Interface	Input	Output	Data format	Protocol	On Error
I1	Image files	Image files	*.JPEG, *.PNG,	Saved to pwd	C2 rejects image
I2	Image file, payload	Encoded image file	*.JPEG, *.PNG, *.TXT	CMD call E1	C2 rejects image
I3	Encoded image files	Encoded image files	*.JPEG, *.PNG	Saved to pwd	C3 rejects image
I4	Encoded image file	Detection results	*.JPEG, *.PNG, *.CSV	CMD call E2	C3 rejects image
I5	Detection results	Comparison results	*.CSV	Saved to pwd	C4 rejects data

Example Payload:

File	Extension	Score
Img_01.jpeg	*.JPEG	33/100
Img_01.png	*.PNG	50/100

Data-flow diagram.



Architecture pattern and justification.

Pattern:

Pipeline, used for the whole system

Fit to the problem:

The experiment runs one step after another.

Team skills:

Each step is in a Python script.

Performance and timing:

No timing requirements.

Scalability:

More images can be added easily.

Hardware constraints:

Runs on anything.

Rejected pattern:

Layered, we don't need separate UI, logic, and storage layers

Decision log.

1. Use a pipeline pattern
 - Alternatives: Layered
 - Why: The experiment runs one step after another, totally unnecessary
2. Write each step in Python.
 - Alternatives: Doing each step by hand.
 - Why: Scripts are more consistent than humans are
3. Use already made tools.
 - Alternatives: Writing our own hiding and detection code
 - Why: Tests tools people actually use
4. Measure a detection score for each format.
 - Alternatives: Binary detection score
 - Why: A score shows how strongly the tool suspects a hidden message